

TEACHER MASTER KEY: HONORS CHALLENGES 1 & 2

Quick-Reference Answers, Coordinates & Debrief Solutions

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Challenge 1: Ship & Drone Kinematic Rendezvous

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Plotted Points on Grid:

- **Ship:** (0s, 300m), (10s, 360m), (20s, 420m), (30s, 480m), (40s, 540m), (50s, 600m), (55s, 630m), (60s, 660m).
- **Drone:** (0 to 20s flat on dock at 0m), (30s, 180m), (40s, 360m), (50s, 540m), (55s, 630m), (60s, 720m).

1. **Rendezvous Coordinates:** Time $t = 55.0$ seconds, Position $x = 630.0$ meters from dock.
2. **Slopes (Speeds):** Ship Slope = $+6.0$ m/s • Drone Slope (after $t=20s$) = $(540 - 0)/(50 - 20) = +18.0$ m/s.
3. **Actual Drone Flight Duration:** $55\text{ s} - 20\text{ s delay} = 35.0$ seconds flying in the air.
4. **Battery Safety Audit:** Total round-trip distance = $630\text{ m outbound} + 630\text{ m return} = 1,260$ meters.
→ Since battery max is $1,200\text{ m}$, **the drone cannot make it back** (crashes 60 m short of the dock).

Challenge 2: Tactical Highway Surveillance Intercept

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Plotted Points on Grid:

- **Target Car:** (0s, 0m) → (20s, 300m) [Leg 1] → (50s, 300m) [Leg 2, flat line] → (90s, 1300m) [Leg 3].
- **Patrol Line A:** Starts at (30s, 0m), connects to rest stop at (50s, 300m).
- **Patrol Line B:** Starts at (30s, 0m), connects to final highway point at (90s, 1300m).

1. **Rest Stop Meaning:** Slope is zero ($\Delta x/\Delta t = 0$), which means velocity is 0 m/s and position does not change (car is stationary).
2. **Cruiser Speed Calculations:**
 - **Strategy A (Rest Stop at $t=50s$):** Slope = $(300\text{ m} - 0\text{ m}) / (50\text{ s} - 30\text{ s}) = 300 / 20 = 15.0\text{ m/s}$ (33.6 mph).
 - **Strategy B (Highway End at $t=90s$):** Slope = $(1,300\text{ m} - 0\text{ m}) / (90\text{ s} - 30\text{ s}) = 1,300 / 60 = 21.67\text{ m/s} \approx 21.7\text{ m/s}$ (48.5 mph).
3. **Speed Limit Constraint (Cruiser Top Speed = 20.0 m/s):**
 - Strategy A requires $15.0\text{ m/s} \leq 20.0\text{ m/s} \rightarrow$ **PHYSICALLY POSSIBLE** ✓
 - Strategy B requires $21.7\text{ m/s} > 20.0\text{ m/s} \rightarrow$ **EXCEEDS SPEED LIMIT / IMPOSSIBLE** ✗→ Therefore, **Strategy A is the only viable option** for the patrol interceptor.